

Stokes' Theorem

Lecture 12

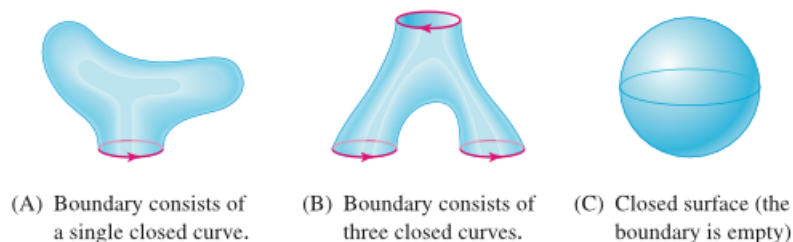
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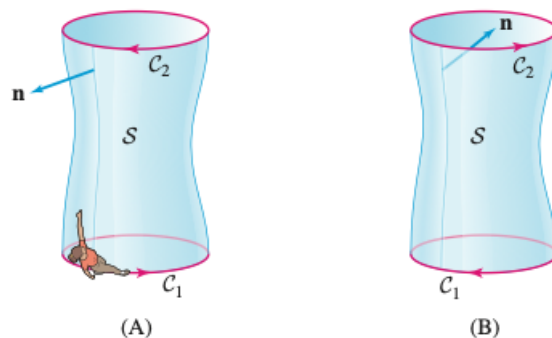
Surface Integrals

Inquiry Question:

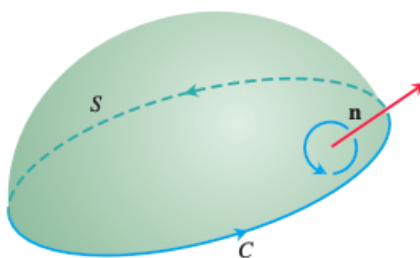
- Idea of circulation in 3—dimension
- **Stokes' Theorem** is an extension of Green's Theorem to three dimensions in which **circulation is related to a surface integral over a surface in $\mathbb{R}^3$** rather than a double integral over a region in the plane).
- The below figures shows three surfaces with different type of boundaries



- Recall from the last session that the orientation of a surface is a continuously varying choice of unit normal vector at each point of a surface.
- When S is oriented, we can specify an orientation of **boundary** called the **boundary orientation**.
- Imagine that you are a unit normal vector along the boundary curve with your head at the head end of the vector and your feet at the tail end.



- For example, the boundary of the surface in Figure above consists of two curves, C_1 and C_2 .
- In (A), the normal vector points to the outside. The woman (representing the normal vector) is walking along C_1 and has the surface to her left, so she is walking in the **positive direction**. The curve C_2 is oriented in the **opposite direction** because she would have to walk along C_2 in that direction to keep the surface on her left.
- The boundary orientations in (B) are reversed because the opposite normal has been selected to orient the surface.
- We require that the surface be **piecewise smooth**, which means that it is a finite union of smooth surfaces joining along smooth curves.
- The orientation of the bounding curve C gives it a right-handed relation to the normal field $\mathbf{n}$. If the thumb of a right hand points along $\mathbf{n}$, the fingers curl in the direction of C .



Stokes' Theorem

1. Let S be a piecewise smooth oriented surface having a piecewise smooth boundary curve C .
2. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be a vector field whose components have continuous first partial derivatives on an open region containing S .

Then the **circulation of $\mathbf{F}$ around C in the direction counter-clockwise with respect to the surface's unit normal vector $\mathbf{n}$** equals the integral of the curl vector field $\vec{\nabla} \times \mathbf{F}$ over S :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma$$

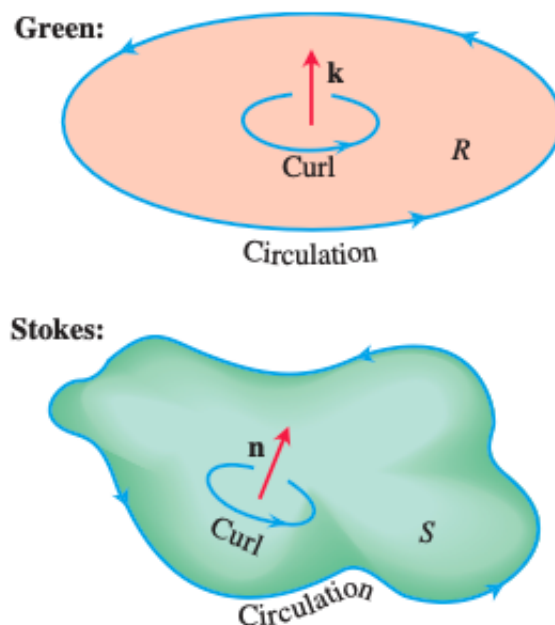


Figure 1: Comparison of Green's Theorem and Stokes' Theorem

If C is a curve in the xy -plane, oriented counterclockwise, and R is the

region in the xy -plane bounded by C , then $d\sigma = dxdy$ and

$$\begin{aligned} (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} &= (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{k} \\ &= \text{kth-component of the curl vector} \\ &= \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \end{aligned}$$

Under these conditions,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma \\ &= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy \end{aligned}$$

which is the [circulation form of the Green's Theorem](#).

The curl measures the extent to which $\mathbf{F}$ fails to be conservative. If $\mathbf{F}$ is conservative, then $\text{curl} F = 0$ and Stokes' Theorem merely confirms what we already know: The circulation of a conservative vector field around a closed path is zero.

Example

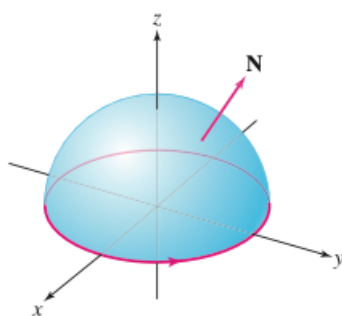
Verify Stokes' Theorem for

$$\mathbf{F}(x, y, z) = \langle -y, 2x, x + z \rangle$$

and the upper hemisphere with outward-pointing normal vectors:

$$\mathcal{S} = \{(x, y, z) : x^2 + y^2 + z^2 = 1, z \geq 0\}$$

given by



Hint: We will show that both the line integral and the surface integral in Stokes' Theorem are equal to 3π .

Verification of Stokes' Theorem

We are to verify Stokes' Theorem for the vector field

$$\mathbf{F}(x, y, z) = \langle -y, 2x, x + z \rangle$$

over the upper hemisphere defined by

$$\mathcal{S} = \{(x, y, z) : x^2 + y^2 + z^2 = 1, z \geq 0\}.$$

According to Stokes' theorem, we need to show that

$$\iint_{\mathcal{S}} \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma = \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Step 1: Compute $\vec{\nabla} \times \mathbf{F}$

The curl of $\mathbf{F}$ is given by:

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & 2x & x+z \end{vmatrix}.$$

Calculating this determinant, we find:

$$\begin{aligned} \vec{\nabla} \times \mathbf{F} = & \hat{i} \left(\frac{\partial}{\partial y}(x+z) - \frac{\partial}{\partial z}(2x) \right) - \hat{j} \left(\frac{\partial}{\partial x}(x+z) - \frac{\partial}{\partial z}(-y) \right) \\ & + \hat{k} \left(\frac{\partial}{\partial x}(2x) - \frac{\partial}{\partial y}(-y) \right). \end{aligned}$$

Calculating each component:

- For $\hat{i}$:

$$\frac{\partial}{\partial y}(x+z) - \frac{\partial}{\partial z}(2x) = 0 - 0 = 0.$$

- For $\hat{j}$:

$$\frac{\partial}{\partial x}(x+z) - \frac{\partial}{\partial z}(-y) = 1 - 0 = 1.$$

- For $\hat{k}$:

$$\frac{\partial}{\partial x}(2x) - \frac{\partial}{\partial y}(-y) = 2 - (-1) = 3.$$

Thus, we have

$$\vec{\nabla} \times \mathbf{F} = \langle 0, -1, 3 \rangle.$$

Step 2: Compute the Surface Integral

The surface integral is given by:

$$\iint_{\mathcal{S}} \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma.$$

1. Parametrization of the Upper Hemisphere

The upper hemisphere can be parametrized in spherical coordinates as follows:

$$\begin{aligned}x &= \sin \phi \cos \theta, \\y &= \sin \phi \sin \theta, \\z &= \cos \phi,\end{aligned}$$

where: - ϕ is the polar angle that varies from 0 to $\frac{\pi}{2}$, - θ is the azimuthal angle that varies from 0 to 2π .

2. Surface Element $d\sigma$ Calculation

To calculate the surface area element $d\sigma$, we will compute the Jacobian of the parametrization.

1. **Tangent Vectors:** The tangent vectors can be obtained by differentiating the position vector $\mathbf{r}(\phi, \theta)$ with respect to ϕ and θ :

$$\mathbf{r}(\phi, \theta) = \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle.$$

- The tangent vector with respect to ϕ is:

$$\mathbf{r}_\phi = \frac{\partial \mathbf{r}}{\partial \phi} = \langle \cos \phi \cos \theta, \cos \phi \sin \theta, -\sin \phi \rangle.$$

- The tangent vector with respect to θ is:

$$\mathbf{r}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = \langle -\sin \phi \sin \theta, \sin \phi \cos \theta, 0 \rangle.$$

2. **Cross Product:** The normal vector $\mathbf{n}$ to the surface can be found using the cross product of the tangent vectors:

$$\mathbf{n} = \mathbf{r}_\phi \times \mathbf{r}_\theta.$$

Computing the cross product gives:

$$\mathbf{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \phi \cos \theta & \cos \phi \sin \theta & -\sin \phi \\ -\sin \phi \sin \theta & \sin \phi \cos \theta & 0 \end{vmatrix}.$$

After calculating the determinant, we find:

$$\mathbf{n} = (\sin^2 \phi \cos \theta, \sin^2 \phi \sin \theta, \sin \phi \cos \phi).$$

3. **Magnitude of the Normal Vector:** The magnitude of the normal vector gives the area scaling factor. The area element $d\sigma$ is given by:

$$d\sigma = |\mathbf{n}| d\phi d\theta.$$

To find the magnitude:

$$|\mathbf{n}| = \sqrt{(\sin^2 \phi \cos \theta)^2 + (\sin^2 \phi \sin \theta)^2 + (\sin \phi \cos \phi)^2}.$$

Simplifying this expression:

$$\begin{aligned} |\mathbf{n}| &= \sqrt{\sin^4 \phi (\cos^2 \theta + \sin^2 \theta) + \sin^2 \phi \cos^2 \phi} \\ &= \sqrt{\sin^4 \phi + \sin^2 \phi \cos^2 \phi} = \sin \phi. \end{aligned}$$

4. **Final Expression for the Surface Element:** Thus, the differential surface area element on the hemisphere becomes:

$$d\sigma = |\mathbf{n}| d\phi d\theta = \sin \phi d\phi d\theta.$$

Computing Surface Integral

Now substituting $d\sigma$ into the integral, we have:

$$\iint_{\mathcal{S}} \nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} (-\sin^2 \phi \sin \theta + 3 \cos \phi \sin \phi) d\phi d\theta.$$

This separates into two integrals:

$$= \int_0^{2\pi} \sin \theta \, d\theta \int_0^{\frac{\pi}{2}} -\sin^2 \phi \, d\phi + 3 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \cos \phi \sin \phi \, d\phi.$$

Evaluating the First Integral:

The first part gives:

$$\int_0^{2\pi} \sin \theta \, d\theta = 0.$$

Thus, this integral contributes 0.

Evaluating the Second Integral:

Now for the second part:

$$3 \cdot \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \cos \phi \sin \phi \, d\phi.$$

Calculating the θ integral first:

$$\int_0^{2\pi} d\theta = 2\pi.$$

Now we evaluate the ϕ integral:

$$\int_0^{\frac{\pi}{2}} \cos \phi \sin \phi \, d\phi.$$

Using the substitution $u = \sin \phi$, $du = \cos \phi \, d\phi$, the limits change as follows:

- When $\phi = 0$, $u = 0$. - When $\phi = \frac{\pi}{2}$, $u = 1$.

Thus, we have:

$$\int_0^{\frac{\pi}{2}} \cos \phi \sin \phi \, d\phi = \int_0^1 u \, du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}.$$

Now substituting back into the integral:

$$3 \cdot 2\pi \cdot \frac{1}{2} = 3\pi.$$

Thus, we have:

$$\iint_{\mathcal{S}} \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma = 3\pi.$$

Step 3: Compute the Line Integral

The boundary $\partial\mathcal{S}$ of the hemisphere is the unit circle in the xy -plane at $z = 0$. The parametrization is:

$$\mathbf{r}(t) = \langle \cos t, \sin t, 0 \rangle, \quad t \in [0, 2\pi].$$

The differential is:

$$d\mathbf{r} = \langle -\sin t, \cos t, 0 \rangle dt.$$

Now we compute:

$$\mathbf{F}(\mathbf{r}(t)) = \langle -\sin t, 2\cos t, \cos t \rangle.$$

The line integral is:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \langle -\sin t, 2\cos t, \cos t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle dt.$$

Calculating the dot product:

$$= \int_0^{2\pi} (\sin^2 t + 2\cos^2 t) dt = \int_0^{2\pi} (1 + \cos^2 t) dt.$$

We can evaluate this integral:

$$\int_0^{2\pi} dt = 2\pi,$$

and

$$\int_0^{2\pi} \cos^2 t \, dt = \pi.$$

Thus,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi + \pi = 3\pi.$$

After verifying both integrals:

$$\iint_S \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma = 3\pi,$$

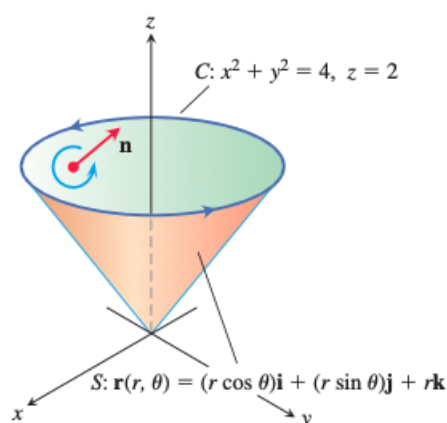
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 3\pi.$$

Example

Find the circulation of the field

$$\mathbf{F} = \langle (x^2 - y), 4z, x^2 \rangle$$

around the curve C in which the plane $z = 2$ meets the cone $z = \sqrt{x^2 + y^2}$, counterclockwise as viewed from above.



Finding the Unit Normal Vector and Surface Element

To compute the unit normal vector and the surface element $d\sigma$ for the parametrization of the cone defined by:

$$\begin{aligned}x &= r \cos \theta, \\y &= r \sin \theta, \\z &= r,\end{aligned}$$

where $0 \leq r \leq 2$ and $0 \leq \theta < 2\pi$.

Step 1: Calculate the Tangent Vectors

To find the unit normal vector, we first need the tangent vectors to the surface parametrized by (r, θ) :

1. **Tangent Vector with respect to r :**

$$\mathbf{r}_r = \frac{\partial \mathbf{r}}{\partial r} = \langle \cos \theta, \sin \theta, 1 \rangle.$$

2. **Tangent Vector with respect to θ :**

$$\mathbf{r}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = \langle -r \sin \theta, r \cos \theta, 0 \rangle.$$

Step 2: Compute the Cross Product

The unit normal vector $\hat{n}$ to the surface can be found using the cross product of the tangent vectors:

$$\hat{n} = \frac{\mathbf{r}_r \times \mathbf{r}_\theta}{|\mathbf{r}_r \times \mathbf{r}_\theta|}.$$

The cross product $\mathbf{r}_r \times \mathbf{r}_\theta$ is computed as follows:

$$\mathbf{r}_r \times \mathbf{r}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta & \sin \theta & 1 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix}.$$

After calculating the determinant, we find:

$$\mathbf{r}_r \times \mathbf{r}_\theta = \langle -r \cos \theta, r \sin \theta, r \rangle.$$

Step 3: Find the Magnitude of the Cross Product

Next, we compute the magnitude of the cross product:

$$|\mathbf{r}_r \times \mathbf{r}_\theta| = \sqrt{(-r \cos \theta)^2 + (r \sin \theta)^2 + r^2}.$$

Calculating this gives:

$$|\mathbf{r}_r \times \mathbf{r}_\theta| = \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta + r^2} = \sqrt{r^2 + r^2} = \sqrt{2r^2} = r\sqrt{2}.$$

Step 4: Compute the Unit Normal Vector

Thus, the unit normal vector is:

$$\hat{n} = \frac{\mathbf{r}_r \times \mathbf{r}_\theta}{|\mathbf{r}_r \times \mathbf{r}_\theta|} = \frac{\langle -r \cos \theta, r \sin \theta, r \rangle}{r\sqrt{2}} = \left\langle -\frac{\cos \theta}{\sqrt{2}}, \frac{\sin \theta}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle.$$

Step 5: Compute the Surface Element $d\sigma$

The differential surface area element $d\Sigma$ on the cone is given by:

$$d\sigma = |\mathbf{r}_r \times \mathbf{r}_\theta| dr d\theta.$$

Substituting the magnitude of the cross product:

$$d\sigma = r\sqrt{2} dr d\theta.$$

Finding Curl of $\mathbf{F}$

The vector field is defined as:

$$\mathbf{F} = \langle x^2 - y, 4z, x^2 \rangle.$$

To find the curl of $\mathbf{F}$, we use the formula:

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix},$$

where: - $F_1 = x^2 - y$, - $F_2 = 4z$, - $F_3 = x^2$.

Calculating the determinant, we get:

$$\begin{aligned} \vec{\nabla} \times \mathbf{F} = & \hat{i} \left(\frac{\partial}{\partial y}(x^2) - \frac{\partial}{\partial z}(4z) \right) - \hat{j} \left(\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial z}(x^2) \right) \\ & + \hat{k} \left(\frac{\partial}{\partial x}(4z) - \frac{\partial}{\partial y}(x^2 - y) \right). \end{aligned}$$

Calculating each component:

1. For $\hat{i}$:

$$\frac{\partial}{\partial y}(x^2) - \frac{\partial}{\partial z}(4z) = 0 - 4 = -4.$$

2. For $\hat{j}$:

$$\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial z}(x^2) = 2x - 0 = 2x.$$

3. For $\hat{k}$:

$$\frac{\partial}{\partial x}(4z) - \frac{\partial}{\partial y}(x^2 - y) = 0 + 1 = 1.$$

Thus, the curl of the vector field is given by:

$$\vec{\nabla} \times \mathbf{F} = \langle -4, -2x, 1 \rangle.$$

In summary, the curl of $\mathbf{F}$ is:

$$\boxed{\vec{\nabla} \times \mathbf{F} = \langle -4, -2x, 1 \rangle}.$$

Thus, in parametrized form, the curl of $\mathbf{F}$ is:

$$\vec{\nabla} \times \mathbf{F} = \langle -4, -2r \cos \theta, 1 \rangle.$$

By using Stokes' Theorem:

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \vec{\nabla} \times \mathbf{F} \cdot \mathbf{n} \, d\sigma \\ &= \int_0^{2\pi} \int_0^2 \frac{1}{\sqrt{2}} (4 \cos(\theta) + r \sin(2\theta) + 1) r \sqrt{2} dr d\theta\end{aligned}$$

First, simplify the integrand by multiplying the constants:

$$\frac{1}{\sqrt{2}} \times \sqrt{2} = 1.$$

Thus, the integral simplifies to:

$$\int_0^{2\pi} \int_0^2 (4 \cos \theta + r \sin(2\theta) + 1) r \, dr \, d\theta.$$

Expanding the terms inside the integral:

$$\int_0^{2\pi} \int_0^2 (4r \cos \theta + r^2 \sin(2\theta) + r) \, dr \, d\theta.$$

Step 2: Integrate with Respect to r

Compute the inner integral with respect to r :

$$\int_0^2 (4r \cos \theta + r^2 \sin(2\theta) + r) \, dr.$$

This integral can be separated into three parts:

$$\int_0^2 4r \cos \theta \, dr + \int_0^2 r^2 \sin(2\theta) \, dr + \int_0^2 r \, dr.$$

Evaluate each part individually:

1. ****First Integral:****

$$\begin{aligned}\int_0^2 4r \cos \theta \, dr &= 4 \cos \theta \int_0^2 r \, dr = 4 \cos \theta \left[\frac{r^2}{2} \right]_0^2 \\ &= 4 \cos \theta \left(\frac{4}{2} \right) = 8 \cos \theta.\end{aligned}$$

2. **Second Integral:**

$$\begin{aligned}\int_0^2 r^2 \sin(2\theta) dr &= \sin(2\theta) \int_0^2 r^2 dr = \sin(2\theta) \left[\frac{r^3}{3} \right]_0^2 \\ &= \sin(2\theta) \left(\frac{8}{3} \right) = \frac{8}{3} \sin(2\theta).\end{aligned}$$

3. **Third Integral:**

$$\int_0^2 r dr = \left[\frac{r^2}{2} \right]_0^2 = \frac{4}{2} = 2.$$

Combining all three results, the inner integral becomes:

$$8 \cos \theta + \frac{8}{3} \sin(2\theta) + 2.$$

Step 3: Integrate with Respect to θ

Now, compute the outer integral:

$$\int_0^{2\pi} \left(8 \cos \theta + \frac{8}{3} \sin(2\theta) + 2 \right) d\theta.$$

This integral can be separated into three parts:

$$8 \int_0^{2\pi} \cos \theta d\theta + \frac{8}{3} \int_0^{2\pi} \sin(2\theta) d\theta + 2 \int_0^{2\pi} d\theta.$$

Evaluate each part individually:

1. **First Integral:**

$$8 \int_0^{2\pi} \cos \theta d\theta = 8 [\sin \theta]_0^{2\pi} = 8(0 - 0) = 0.$$

2. **Second Integral:**

$$\begin{aligned}\frac{8}{3} \int_0^{2\pi} \sin(2\theta) d\theta &= \frac{8}{3} \left[-\frac{\cos(2\theta)}{2} \right]_0^{2\pi} = \frac{8}{3} \left(-\frac{\cos(4\pi) - \cos(0)}{2} \right) \\ &= \frac{8}{3} \left(-\frac{1 - 1}{2} \right) = 0.\end{aligned}$$

3. **Third Integral:**

$$2 \int_0^{2\pi} d\theta = 2 [\theta]_0^{2\pi} = 2(2\pi - 0) = 4\pi.$$

Combining all three results, the outer integral becomes:

$$0 + 0 + 4\pi = 4\pi.$$

Step 4: Final Result

Therefore, the value of the double integral is:

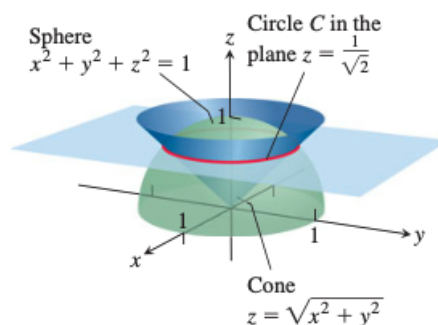
$$\boxed{4\pi}.$$

Example

Calculate the circulation of the vector field

$$\mathbf{F}(x, y, z) = \langle x^2 + z, y^2 + 2x, z^2 - y \rangle$$

along the curve of intersection of the sphere $x^2 + y^2 + z^2 = 1$ with the cone $z = \sqrt{x^2 + y^2}$ traversed in the counter-clockwise direction around the z-axis when viewed from above.



Step 1: Identify the Curve C

The curve C is the intersection of the sphere and the cone. Solving the equations simultaneously:

$$\begin{cases} x^2 + y^2 + z^2 = 1 & (\text{Sphere}) \\ z = \sqrt{x^2 + y^2} & (\text{Cone}) \end{cases}$$

Substituting $z = \sqrt{x^2 + y^2}$ into the sphere equation:

$$\begin{aligned} x^2 + y^2 + (\sqrt{x^2 + y^2})^2 &= 1 \\ x^2 + y^2 + x^2 + y^2 &= 1 \\ 2(x^2 + y^2) &= 1 \\ x^2 + y^2 &= \frac{1}{2} \end{aligned}$$

Thus, C is a circle of radius $r = \frac{1}{\sqrt{2}}$ in the plane $z = \frac{1}{\sqrt{2}}$.

Parametrization of C :

$$\begin{aligned} x &= \frac{1}{\sqrt{2}} \cos \theta, \\ y &= \frac{1}{\sqrt{2}} \sin \theta, \\ z &= \frac{1}{\sqrt{2}}, \end{aligned}$$

where $\theta \in [0, 2\pi)$.

Step 2: Apply Stokes' Theorem

Stokes' Theorem states:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} d\sigma,$$

where:

- S is a surface bounded by C ,
- $\mathbf{n}$ is the unit normal vector to S ,
- $d\sigma$ is the differential surface area element.

We choose S to be the disk in the plane $z = \frac{1}{\sqrt{2}}$ bounded by C .

Step 3: Compute the Curl of $\mathbf{F}$

Given:

$$\mathbf{F}(x, y, z) = \langle x^2 + z, y^2 + 2x, z^2 - y \rangle$$

The curl of $\mathbf{F}$ is:

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + z & y^2 + 2x & z^2 - y \end{vmatrix}$$

Calculating each component:

1. For $\hat{i}$:

$$\frac{\partial}{\partial y}(z^2 - y) - \frac{\partial}{\partial z}(y^2 + 2x) = (-1) - (0) = -1$$

2. For $\hat{j}$:

$$\frac{\partial}{\partial z}(x^2 + z) - \frac{\partial}{\partial x}(z^2 - y) = 1 - 0 = 1$$

3. For $\hat{k}$:

$$\frac{\partial}{\partial x}(y^2 + 2x) - \frac{\partial}{\partial y}(x^2 + z) = 2 - 0 = 2$$

Thus, the curl is:

$$\vec{\nabla} \times \mathbf{F} = \langle -1, 1, 2 \rangle$$

Step 4: Determine the Unit Normal Vector $\vec{n}$

Since S lies in the plane $z = \frac{1}{\sqrt{2}}$, the unit normal vector pointing upwards is:

$$\vec{n} = \langle 0, 0, 1 \rangle$$

Step 5: Compute the Surface Integral

The differential surface area element $d\sigma$ for the disk S in polar coordinates is:

$$d\sigma = r \, dr \, d\theta$$

Thus, the surface integral becomes:

$$\iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \langle -1, 1, 2 \rangle \cdot \langle 0, 0, 1 \rangle r \, dr \, d\theta = \iint_S 2r \, dr \, d\theta$$

Step 6: Evaluate the Integral

Set up the integral with the given limits:

$$\iint_S 2r \, dr \, d\theta = 2 \int_0^{2\pi} \int_0^{\frac{1}{\sqrt{2}}} r \, dr \, d\theta$$

Compute the inner integral (with respect to r):

$$\int_0^{\frac{1}{\sqrt{2}}} r \, dr = \left[\frac{r^2}{2} \right]_0^{\frac{1}{\sqrt{2}}} = \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

Now, compute the outer integral (with respect to θ):

$$2 \int_0^{2\pi} \frac{1}{4} \, d\theta = \frac{1}{2} \int_0^{2\pi} d\theta = \frac{1}{2} \times 2\pi = \pi$$

Step 7: Conclusion via Stokes' Theorem

By Stokes' Theorem:

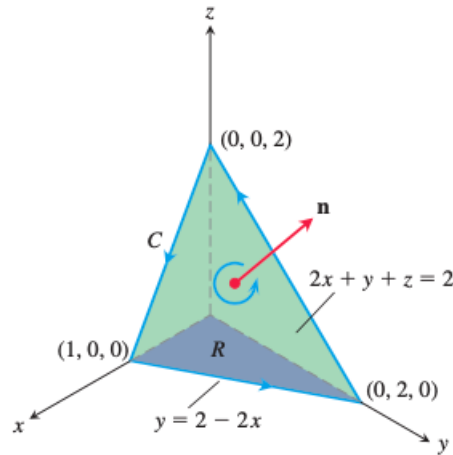
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma = \pi$$

Thus, the circulation of the vector field $\mathbf{F}$ around the curve C is:

$$\boxed{\pi}$$

Examples

Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, if $\mathbf{F} = \langle xz, xy, 3xz \rangle$ and C is the boundary of the portion of the plane $2x + y + z = 2$ in the first octant, traversed counter-clockwise as viewed from above.



We will use **Stokes' Theorem** to evaluate the line integral. Stokes' Theorem relates a line integral over a closed curve C to a surface integral over a surface S bounded by C :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} d\sigma,$$

where $\mathbf{n}$ is the unit normal vector to the surface S .

1. Compute the Curl of $\mathbf{F}$

Given:

$$\mathbf{F} = \langle xz, xy, 3xz \rangle$$

The curl of $\mathbf{F}$ is:

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xy & 3xz \end{vmatrix}$$

Calculating each component:

$$\begin{aligned} (\vec{\nabla} \times \mathbf{F})_i &= \frac{\partial(3xz)}{\partial y} - \frac{\partial(xy)}{\partial z} = 0 - 0 = 0, \\ (\vec{\nabla} \times \mathbf{F})_j &= \frac{\partial(xz)}{\partial z} - \frac{\partial(3xz)}{\partial x} = x - 3z, \\ (\vec{\nabla} \times \mathbf{F})_k &= \frac{\partial(xy)}{\partial x} - \frac{\partial(xz)}{\partial y} = y - 0 = y. \end{aligned}$$

Thus,

$$\vec{\nabla} \times \mathbf{F} = \langle -x, x - 3z, y \rangle.$$

However, on the plane $z = 2 - 2x - y$, the curl of the vector field is:

$$\vec{\nabla} \times \mathbf{F} = \langle 0, 7x + 3y - 6, y \rangle.$$

2. Parameterize the Surface S

The plane is given by:

$$2x + y + z = 2.$$

Solving for z :

$$z = 2 - 2x - y.$$

Since the surface lies in the first octant, the bounds for x and y are:

$$0 \leq x \leq 1, \quad 0 \leq y \leq 2 - 2x.$$

3. Determine the Normal Vector $\mathbf{n}$

The plane $2x + y + z = 2$ has a normal vector:

$$\mathbf{n} = \langle 2, 1, 1 \rangle.$$

The unit normal vector is then

$$\mathbf{n} = \frac{1}{\sqrt{6}} \langle 2, 1, 1 \rangle.$$

The differential surface element is:

$$d\sigma = \sqrt{6} \, dy \, dx.$$

4. Set Up the Surface Integral

Applying Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma.$$

Substitute $\vec{\nabla} \times \mathbf{F}$ and $\mathbf{n}$:

$$\iint_S (\vec{\nabla} \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma = \int_0^1 \int_0^{2-2x} [0 \cdot 2 + (7x + 3y - 6) \cdot 1 + y \cdot 1] \, dy \, dx.$$

Simplify the integrand:

$$= \int_0^1 \int_0^{2-2x} (7x + 4y - 6) \, dy \, dx.$$

5. Evaluate the Integral

First, integrate with respect to y :

$$\int_0^{2-2x} (7x + 4y - 6) \, dy = 7x \int_0^{2-2x} dy + 4 \int_0^{2-2x} y \, dy - 6 \int_0^{2-2x} dy.$$

Calculate each integral:

$$\begin{aligned} 7x \int_0^{2-2x} dy &= 7x(2 - 2x), \\ 4 \int_0^{2-2x} y \, dy &= 4 \left[\frac{1}{2} y^2 \right]_0^{2-2x} = 2(2 - 2x)^2, \\ -6 \int_0^{2-2x} dy &= -6(2 - 2x). \end{aligned}$$

Combine the results:

$$= 7x(2 - 2x) + 2(2 - 2x)^2 - 6(2 - 2x).$$

Expand and simplify:

$$\begin{aligned} &= 14x - 14x^2 + 2(4 - 8x + 4x^2) - 12 + 12x \\ &= 14x - 14x^2 + 8 - 16x + 8x^2 - 12 + 12x \\ &= (-14x^2 + 8x^2) + (14x - 16x + 12x) + (8 - 12) \\ &= -6x^2 + 10x - 4. \end{aligned}$$

Now, integrate with respect to x :

$$\int_0^1 (-6x^2 + 10x - 4) \, dx = \left[-2x^3 + 5x^2 - 4x \right]_0^1.$$

Substitute the limits:

$$= (-2(1)^3 + 5(1)^2 - 4(1)) - (0) = -2 + 5 - 4 = -1.$$

6. Conclusion

The value of the line integral is:

$$\boxed{-1}$$

Stokes Theorem for Surfaces with Holes

Stokes' Theorem holds for an oriented surface S that has one or more holes. The surface integral over S of the normal component of $\vec{\nabla} \times \mathbf{F}$ equals the sum of the line integrals around all the boundary curves of the tangential component of $\mathbf{F}$, where the curves are to be traced in the direction induced by the orientation of S . For such surfaces the theorem is unchanged, but C is considered as a union of simple closed curves.

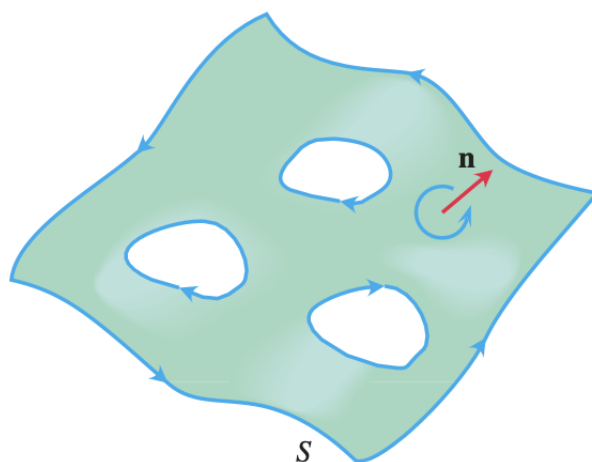


Figure 2: Stokes' Theorem also holds for oriented surfaces with holes. Consistent with the orientation of S , the outer curve is traversed counterclockwise around $\mathbf{n}$ and the inner curves surrounding the holes are traversed clockwise.